Type MS · Evaluate kidney transplants and dialysis as treatments

Answer 1 of 5 · Question 8 part whole · 5 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

Number			
8	Any five from the following: **Kidney transplants - advantages** • patient can lead a normal life; • less cost (overall) to patient/health service; • no need to monitor diet; **Kidney transplants - disadvantages** • shortage of organ donors; • requires surgery with the risk of infection; • chance of rejection; • patient must take (immune-suppressant) medication for life; **Dialysis - advantages** • prolongs life temporarily/temporary solution; • no need for (immune-suppressant) medication; • available to all patients; **Dialysis - disadvantages** • regular visits (to hospital) for dialysis/long time spent on dialysis machine/need frequent treatment; • diet needs to be monitored; • less cost effective (justified);		5

Total marks 5

Type MS · Evaluate kidney transplants and dialysis as treatments

Answer 1 of 5 · Question 8 part whole · 5 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Evaluate kidney transplants and dialysis as treatments

Answer 2 of 5 · Question 4 part b · 5 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

4(a)(i)	A protein;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	A the molecules are too large;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(iii)	• more glucose in blood/filtrate; • as little/no insulin to lower blood glucose levels/glucose not converted to glycogen; • not all/too much to be reabsorbed; • some found in urine;		**(3)**

Question Number	Answer	Additionalguidance	Mark
4(a)(iv)	$\frac{25 \times 100}{53}$; 47%;	Full marks for correct final answer Allow 47.2% / 47.17 Ecf allow one mark for correct answer using incorrect figures in calculation	**(2)**

Type MS · Evaluate kidney transplants and dialysis as treatments

Answer 2 of 5 · Question 4 part b · 5 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
4(b)	Any five from: Advantages • transplant permanent/carried out once; • can return to normal life; • not connected to machine for long periods/no need for frequent hospital visits; Disadvantages • risk of rejection; • risk of infection (during/after surgery)/reduced immunity; • difficulties in finding donor/long waiting lists for donor;		**(5 max)**

Total 12 Marks

Type MS · Evaluate kidney transplants and dialysis as treatments

Answer 2 of 5 · Question 4 part b · 5 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

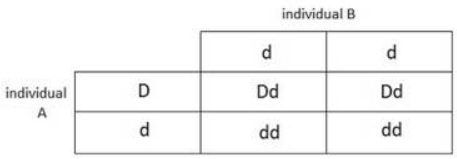
Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Evaluate kidney transplants and dialysis as treatments

Answer 3 of 5 · Question 7 part b.i · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
7	(a)	Any three from: • changes the DNA/base/codon sequence; • changes the order of the amino acids/different amino acid coded for; • changes the function/shape of the protein/different protein; • affects a characteristic or function/genotype in the individual;		3
	(b) (i)	Two from: • dialysis has to be completed several times a week/transplant permanent; • increased life expectancy; • less restrictions on diet; • better quality of life/freedom to move around/normal lifestyle;		2
	(ii)	100 x 3233÷4733; = 68.3/68		2
	(iii)	Two from: • people have two kidneys; • two recipients from one donor; • living donation/can live with just one;		2
	(c) (i)	0.5:1;	Accept 1:2	1
	(ii)	 • correct genotype individual A; • correct genotype individual B; • 50% offspring heterozygous/correct genotype and affected; • 50% homozygous recessive/correct genotype and not affected;		4

Total 14 marks

Type MS · Evaluate kidney transplants and dialysis as treatments

Answer 3 of 5 · Question 7 part b.i · 2 marks

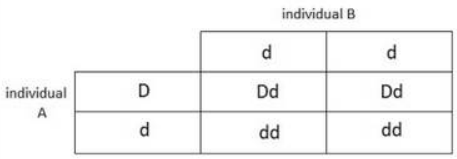
2023 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · Evaluate kidney transplants and dialysis as treatments

Answer 4 of 5 · Question 7 part b.iii · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
7	(a)	Any three from: • changes the DNA/base/codon sequence; • changes the order of the amino acids/different amino acid coded for; • changes the function/shape of the protein/different protein; • affects a characteristic or function/genotype in the individual;		3
	(b) (i)	Two from: • dialysis has to be completed several times a week/transplant permanent; • increased life expectancy; • less restrictions on diet; • better quality of life/freedom to move around/normal lifestyle;		2
	(ii)	100 x 3233÷4733; = 68.3/68		2
	(iii)	Two from: • people have two kidneys; • two recipients from one donor; • living donation/can live with just one;		2
	(c) (i)	0.5:1;	Accept 1:2	1
	(ii)	 • correct genotype individual A; • correct genotype individual B; • 50% offspring heterozygous/correct genotype and affected; • 50% homozygous recessive/correct genotype and not affected;		4

Total 14 marks

Type MS · Evaluate kidney transplants and dialysis as treatments

Answer 4 of 5 · Question 7 part b.iii · 2 marks

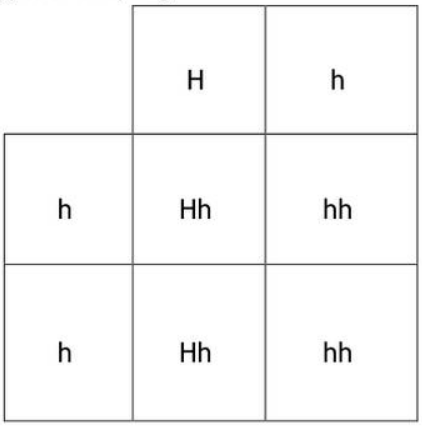
2023 JAN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

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Type MS · Evaluate kidney transplants and dialysis as treatments

Answer 5 of 5 · Question 3 part c · 2 marks

2024 JUN · Paper 02 (regional R) · find part c in this mark scheme extract

3 (a)	change in the order of bases in DNA;	Allow a description of a type of mutation	(1)
(b)	parent gametes; genotypes of offspring;  	Allow genetic cross Also allow different orientation of alleles in Punnett Square	(2)
(c)	**Advantages** • no (costly) visits to hospital; • no inconvenience of dialysis; • improved quality of life/ longer life/long term solution; **Disadvantages** • reference to difficulty in finding a donor; • rejection/reference to anti-rejection medication for life; • reference to risks associated with surgery;	any one from advantages and any one from disadvantages	(2)

Total for Question 3 = 5 marks